

EDUCATION

Carnegie Mellon University <i>M.S. in Software Engineering</i> GPA: 4.0 / 4.0 Selected Coursework: Foundations of Software Engineering, Software Refactoring, Distributed Systems	Expected May 2027
University of Wisconsin–Madison <i>B.S. in Computer Science & B.S. in Computer Engineering</i> GPA: 3.72 / 4.0 Selected Coursework: Software Engineering, Algorithms, Operating Systems	May 2024

WORK EXPERIENCE

NewsBreak <i>Software Engineering Intern</i>	May – August 2026 <i>Mountain View, CA</i>
- Built SportsBreak’s Python FastAPI / MongoDB backend from its first commit to 89 REST endpoints with ~1,450 fully offline tests; added a ruff / pyright / pytest GitHub Actions gate after tests sat red on main. - Stood up the service on production AWS EKS (Docker, Helm, Jenkins) on day 3 and migrated nine third-party integrations server-side so the iOS binary ships zero third-party API credentials. - Wrote tests for an AI-agent-written catalog cache, then redesigned it when pod memory neared its 512 MiB limit: 13.2 MB keyed per user (zero hit rate) → 10.9 MB shared, 78× smaller. - Spot-checked two closed security tickets against the code, found the fixes absent, and reopened them with evidence; wrote 34 contract-first API specs and ~10,000 lines of engineering docs. - Designed an LLM news-alert pipeline that scores each article on arrival and fires only above its topic’s rolling 7-day percentile; calibrated on 4,100+ shadow decisions, it cut filler 34% → 4%. - Sole SwiftUI engineer; shipped SportsBreak to the App Store (~6,000 installs, 1,119 peak daily active users). - Traced another team’s 65% LLM failure rate via Langfuse traces to an unbounded reasoning loop (p95 1,197 tokens vs. 16,384 runaways) and sized a 2,048-token cap against it.	
Haddee Education <i>Engineering Lead</i>	June – September 2025 <i>San Diego, CA</i>
Led an 8-intern team through Agile sprints to ship a React, Next.js, FastAPI, and Supabase web app on GCP App Engine; led PR reviews and mentored interns on Git workflows and testing protocols.	
ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch <i>Full Stack Developer</i>	September – December 2023 <i>Beijing, China</i>
Built a VC-backed startup’s LLM mock-interview web app from 0→1 as the sole engineer in React, Next.js, Supabase, and PostgreSQL; iterated with clients and won contracts with mid-sized companies.	

RESEARCH EXPERIENCE

Control & Learning Group — Carnegie Mellon University <i>Graduate Researcher</i> Advisor: Prof. Yorie Nakahira	February – June 2026
- Developed an Online Adaptive Probabilistic Safety Certificate (PSC) with PhD researchers for stochastic systems; outperformed Adaptive MPC in 3-DOF vehicle simulations under environmental uncertainty. - Implemented Bayesian estimators for real-time road-friction adaptation (LuGre tire models) and mapped natural-language instructions to formal safety specs with multi-turn LLMs.	

SKILLS

Languages	Python, Java, JavaScript, TypeScript, CSS, HTML, SQL, Go, Swift, C, Bash
Practices	Git, CI/CD, GitHub Actions, Unit Testing, pytest, TDD, REST APIs, Agile/Scrum, Jira
Cloud & Infra	AWS, GCP, Kubernetes (EKS), Docker, Helm, Jenkins, Linux, Grafana, Langfuse
Frameworks	FastAPI, Pydantic, React, Next.js, SwiftUI, LangChain, Supabase
Data & ML	MongoDB, PostgreSQL, MySQL, PyTorch, CUDA, RAG

PROJECTS

- my-av:** Built a PyTorch CNN-GRU that predicts steering and speed from 20-frame (1s) video; built a data pipeline over 33 hours of Comma2k19 and ran streaming inference on local CUDA and GCP Vertex AI.
- Wisconsin Autonomous:** Integrated wheel-speed encoders on the competition rover and streamed real-time data to the onboard NVIDIA Jetson.